Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	14 October 2026
Team ID	
Project Name	Metro Ticket Generating System
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Ticket Booking (Service Catalog)	Book through the 'Book A Metro Ticket' catalog item on the Service Portal Select Starting From and Going To stations (reference to Metro Station's Details) Select Type of journey (Single / Return) and No of Passengers (1–4)
FR-2	Station Master Data	Store stations in the custom table Metro Station's Details (u_metro_station_details) Station Name used as display value in the form
FR-3	Automatic Fare Calculation	Catalog client script calculates fare = route fare × passengers Return journey = double fare; amount fields are read-only
FR-4	Payment Mode Selection	Select UPI, Card or Others Catalog UI policy shows a mandatory 'Enter payment mode' field when Others is selected
FR-5	QR Code Ticket Generation	onSubmit client script builds a QR image (QR Code API) QR shown in the Metro QR Widget popup; second Order Now places the order
FR-6	Booking Storage & Verification	Request (REQ) and Requested Item (RITM) created on submission Flow Designer stores the booking in Metro Ticket Booking table and creates a catalog task for Station Managers

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Simple Service Portal form with 8 fields; fare fills automatically; works in any desktop or mobile browser.
NFR-2	**Security**	Default ACLs on custom tables – passengers can book through the catalog but cannot open the database tables; flow runs as System.

NFR-3	**Reliability**	Mandatory fields and read-only fare fields prevent incomplete or wrong bookings; every request gets a unique REQ/RITM number.
NFR-4	**Performance**	Fare appears instantly in the browser; QR popup appears within seconds; flow completes immediately after submission.
NFR-5	**Availability**	Hosted on the ServiceNow cloud (PDI) – available 24×7 through the internet.
NFR-6	**Scalability**	New stations and routes can be added as table records and fare entries; new features (payment gateway, AFC gates) can be added on the same platform.